



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
29-11-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

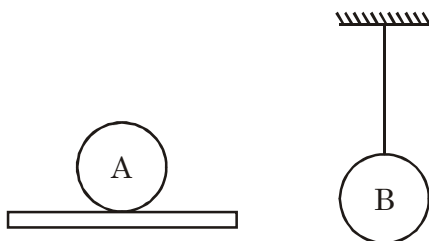
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A uniform rod of mass m , length ℓ , area of cross-section A and Young's modulus Y is pulled by a force $F = Mg$ on a rough table ($\mu = 0.5$). Choose the **INCORRECT** option(s).
- (A) Strain at mid point of rod is $\frac{mg}{4AY}$.
- (B) Elastic potential energy density at mid point is $\frac{m^2 g^2}{16A^2 Y}$.
- (C) Acceleration of rod is $g/2$.
- (D) Stress at distance $\ell/4$ from end, where force is applied, is $\frac{3mg}{4A}$.

Space for Rough Work

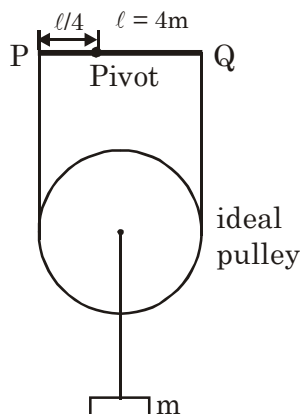
2. An escalator moving downward, a boy of mass 40 kg walks on escalator from bottom to top, with speed one and half times the speed of escalator with respect to escalator. If height difference between top and bottom of escalator is 12 m and escalator moving down with constant speed 6 m/s. Choose the **CORRECT** statement(s) :
- (A) Work done by boy to move from bottom to top is 14.4 kJ.
 (B) Work done by boy on escalator is zero.
 (C) Speed of boy w.r.t ground is 3 m/s.
 (D) Work done by boy to move from bottom to top 4.8 kJ.
3. Two identical iron sphere, one-(A) kept on thermally insulated stand, other (B) suspended by an insulating rigid rod. Both spheres are given equal amount of heat. Then



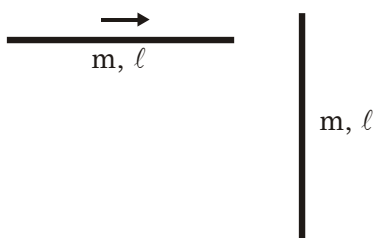
- (A) Temperature of both spheres increases by equal amount.
 (B) Gravitational potential energy of sphere-(A) increases.
 (C) Centre of mass of sphere-(B) shifts downward.
 (D) Increase in temperature of sphere-(B) is greater than (A).

Space for Rough Work

4. A stick of mass m , length ℓ is pivoted at distance $\ell/4$ from end P. An ideal pulley is hanged by means of light string. Another block of mass m is connected by light string to the centre of pulley. System is released from rest. Select the **CORRECT** options :-



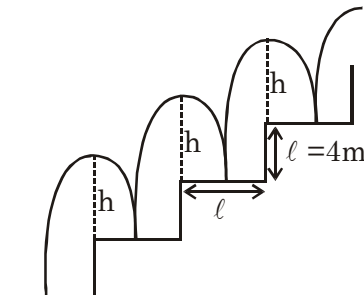
- (A) Just after release acceleration of block is 6 m/s^2 .
 (B) Tension in the string on the left and right side of pulley is different.
 (C) Angular acceleration of stick just after release is 6 rad/sec^2 .
 (D) Stick rotates anticlockwise just after release.
5. A uniform rod of mass m , length ℓ is moving with speed V on a smooth horizontal surface and strikes another identical rod at one end, and sticks to it, and form a right angled object. Choose correct option(s).



- (A) Angular velocity of object after collision is $\frac{3V}{5\ell}$.
 (B) Instantaneous axis of rotation is at $\frac{5\ell}{6}$ from COM of system just after collision.
 (C) Loss in kinetic energy of system is $\frac{7}{40}mv^2$.
 (D) Velocity of COM of system after collision is V .

Space for Rough Work

6. A ball bounces on a staircase, such that it strikes at each step at the same point (midpoint of step) and also bounces to same height above each step as shown in the figure. Co-efficient of restitution between ball & steps is $e = \frac{2}{3}$. Then



- (A) Vertical component of velocity just after each collision is 8 m/s.
 (B) Bouncing height h is 4m.
 (C) Ball strikes at each step with same speed.
 (D) Time interval between two consecutive collisions is 2 sec.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

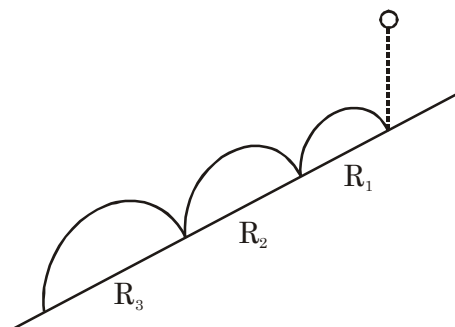
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										-	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A ball is dropped on an inclined plane, it strikes the plane elastically. After each collision it's range is R_1 , R_2 and R_3 respectively. If $R_2 = \frac{a(R_1 + R_3)}{6.5}$, find a.



Space for Rough Work

2. A uniform chain is hanged by two hinges 1m above ground as shown in figure-1. Now its middle point is pulled down, so that its becomes straight as in figure-2. During pulling downward 0.4 Joules work is done. Mass of chain is 2 kg and length is 2m. Find initial height of COM (in meters) from ground in figure-1.

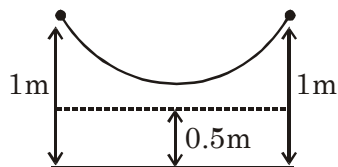


Figure-1

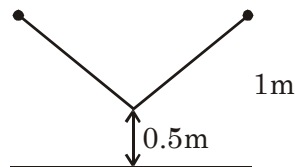
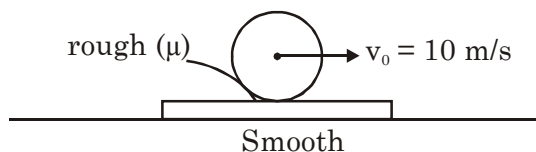


Figure-2

3. 12 gm ice at -20° put in 30 gm water at 20°C . Find amount of ice (in gm) remaining at equilibrium ($S_{\text{ice}} = 0.5 \text{ cal/gm}^\circ\text{C}$, $L_{\text{ice}} = 80 \text{ cal/gm}$, $S_{\text{w}} = 1 \text{ cal/gm}^\circ\text{C}$)

Space for Rough Work

4. A cylinder of mass m is given velocity v_0 without rotation on a long plank of same mass. There is no friction between plank and ground, top surface of plank is rough. What will be the velocity (in m/s) of cylinder when it starts rolling without slipping on the plank ?



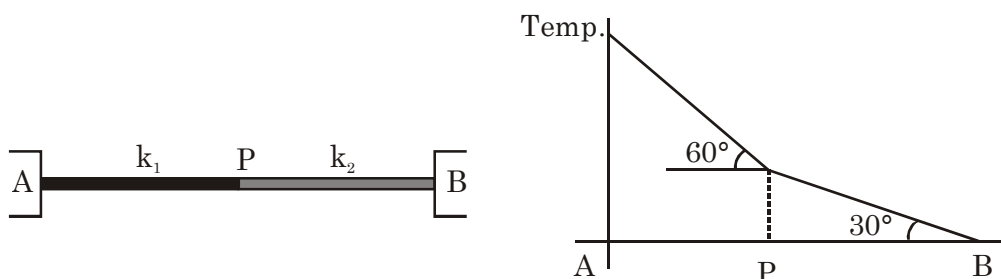
5. A bomb is projected vertically upward from ground. At highest point it explodes in two parts having mass ratio 1 : 2. Both the part land on ground at equal distance from point of projection at an interval of 4 sec. If both parts land on ground at time t_1 and t_2 after explosion, find $\frac{t_1 + t_2}{5}$.
6. A spherical black body at absolute temperature T is surrounded by a thin spherical and concentric shell of radius R , black on both sides. Surrounding temperature is constant T_0 . Find the ratio of rate of energy loss by black body before and after it is surrounded by spherical shell, consider the space between shell and black body is evacuated with no conduction loss (given $R = 2m$, $r = 1m$)

Space for Rough Work

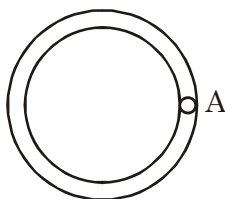
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Two conducting rods are joined at P and end A and end B are kept at fixed temperature. Temperature variation from A to B is given below (k_1, k_2 conductivity of rods). Find ratio of thermal conductivities $\left(\frac{k_2}{k_1}\right)$ of the rods.

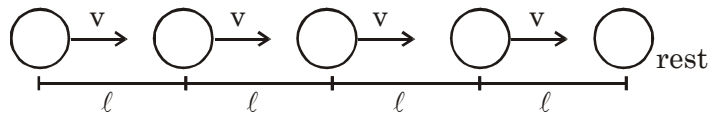


2. A thin ring of volume mass density ρ , cross-sectional area A , radius R and young's modulus of elasticity E , is rotated about an axis passing through its centre with angular velocity ω . If increases in circumference (assumed small) is $\frac{18\pi R^3 \rho \omega^2}{aE}$. Find a .

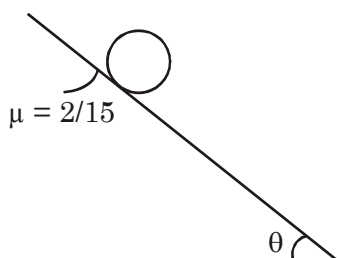


Space for Rough Work

3. 5 identical balls are at rest on a smooth horizontal table. Four balls are given velocity v and the last one is at rest. Each collision is perfectly inelastic from the shown instant. If total time till last collision is $\frac{2.5 a \ell}{v}$, find a

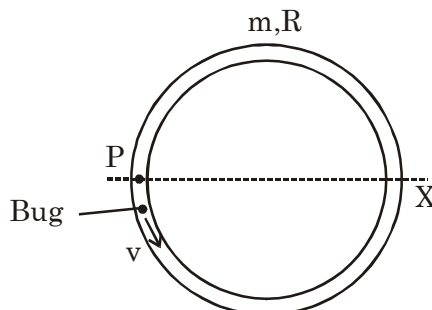


4. A cylinder is released from rest on a rough inclined plane. Maximum value of angle θ for which cylinder rolls without slipping after the release is $\tan^{-1}\left(\frac{\alpha}{\beta}\right)$. Find $(\alpha + \beta)$.

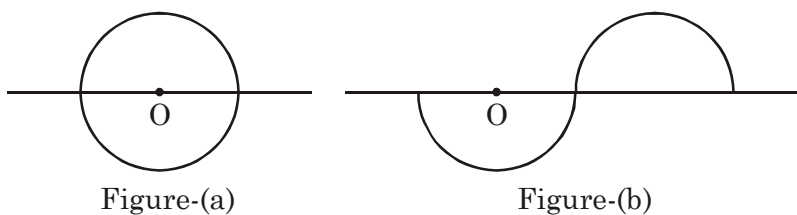


Space for Rough Work

5. A thin ring of mass (100 gm) and radius R (3 m) is pivoted at point P. A bug of mass (50 gm) starts moving from point P along the circumference of ring with speed V (8 m/s) w.r.t to ring. Find speed (in m/s) of bug w.r.t ground when it reaches at diametrically opposite point X of point P. Bug is moving with constant speed w.r.t. ring.



6. A uniform disc is cut along its diameter and shifted towards right as shown in figure-(b). Moment of inertia of discs about an axis passing through centre O and perpendicular to the plane is I_2 , in figure-(b) and I_1 in figure-(a). If $\frac{I_2}{I_1} = \alpha$, find α .



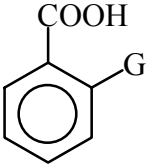
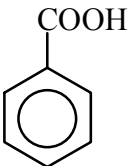
Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I : (Maximum Marks: 24)**

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-
1. The **CORRECT** order of increasing bond angles in the following is/are.
(A) $\text{ClO}_4^- < \text{ClO}_2^- < \text{ClO}_3^-$ (B) $\text{NO}_2^\ominus < \text{NO}_3^\ominus < \text{NO}_2^+$
(C) $\text{SbH}_3 < \text{AsH}_3 < \text{PH}_3 < \text{NH}_3$ (D) $\text{SnCl}_4 < \text{GeCl}_4 < \text{SiCl}_4 < \text{CCl}_4$
 2. A mixture of C_4H_8 and C_2H_4 was completely burnt in excess of oxygen yielding equal volumes of CO_2 and steam. Calculate the percentage of the compounds in the original mixture :
(A) 25% C_4H_8 and 75% C_2H_4 (B) 30% C_4H_8 and 70% C_2H_4
(C) 75% C_4H_8 and 25% C_2H_4 (D) 50% C_4H_8 and 50% C_2H_4
-

Space for Rough Work

3.  is more acidic than  , if G is/are -
- (A) $-\text{OH}$ (B) $-\text{NO}_2$ (C) $-\text{COOH}$ (D) $-\text{CH}_3$
4. In which of the following molecule(s), even number of total lone pairs are present.
- (A) N_3^- (B) XeO_4
 (C) Thionyl chloride (SOCl_2) (D) Sulphuryl chloride (SO_2Cl_2)
5. In the reaction $\text{Ca} + \text{H}_2 \rightarrow \text{CaH}_2$
 Select the incorrect statement -
- (A) Calcium undergoes oxidation (B) Hydrogen undergoes reduction
 (C) Calcium act as oxidising agent (D) Hydrogen act as oxidising agent
6. What is the incorrect IUPAC name of the following :
- $\text{CH}_3-\text{CH}=\text{CH}-\underset{\text{CN}}{\text{CH}}-\text{CH}_2-\text{CH}_3$
- (A) 4-Cyano hex-2-ene (B) Hex-4-ene-3-nitrile
 (C) 2-Ethyl pent-3-ene nitrile (D) 2-(prop-1-enyl) butane nitrile

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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+									
●	●	0	0	0	0	0	0	0	0
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2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Consider the following molecules/ion, calculate (x . y + z)

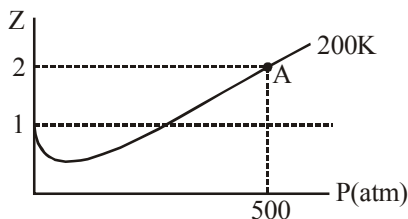
CO, CN^o and N₂

here, x = sum of total number of σ bonds in given three species

y = sum of total number of π bonds in given three species

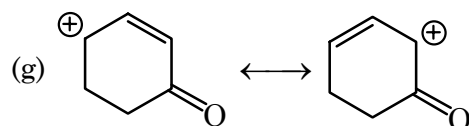
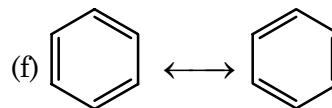
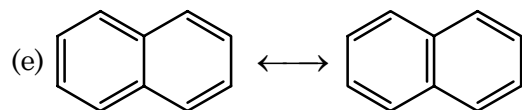
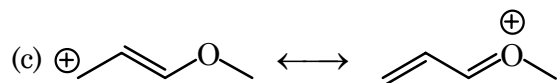
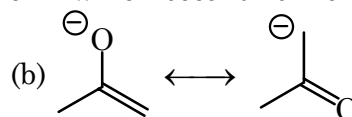
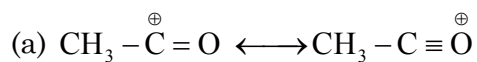
z = sum of total number of lone pairs in given three species

2. 'Z' v/s 'P' graph is plotted for 1 mole of hypothetical gas, Volume of gas (in ml) at this point 'A' is [R = 0.08 atm L/mole-K]



Space for Rough Work

3. Total number of pairs of resonating structure in which second is more stable than first.

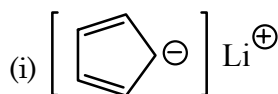
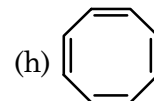
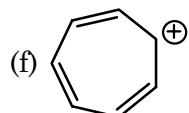
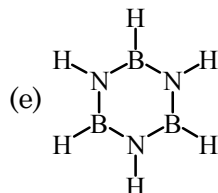
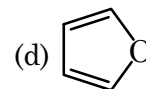
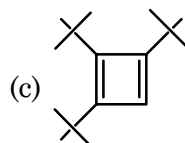
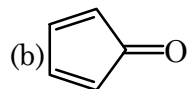
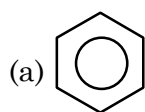


4. The permissible value of m if changed as $-2\ell, \dots, 0, \dots, +2\ell$ (all integral values) and all the other permissible value does not change.

Considering the above change the maximum number of electrons in 3d subshell?

Space for Rough Work

5. The number of Balmer lines when electron(s) deexcite from 6th excited state to ground state.
6. How many compounds are aromatic nature

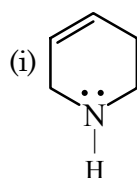
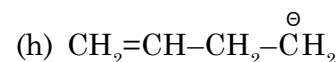
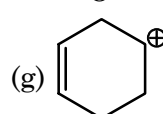
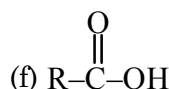
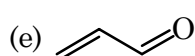
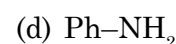
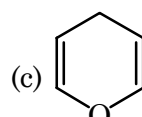
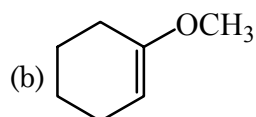
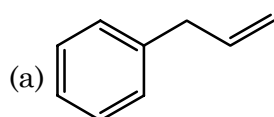


Space for Rough Work

SECTION-III : (Maximum Marks : 18)

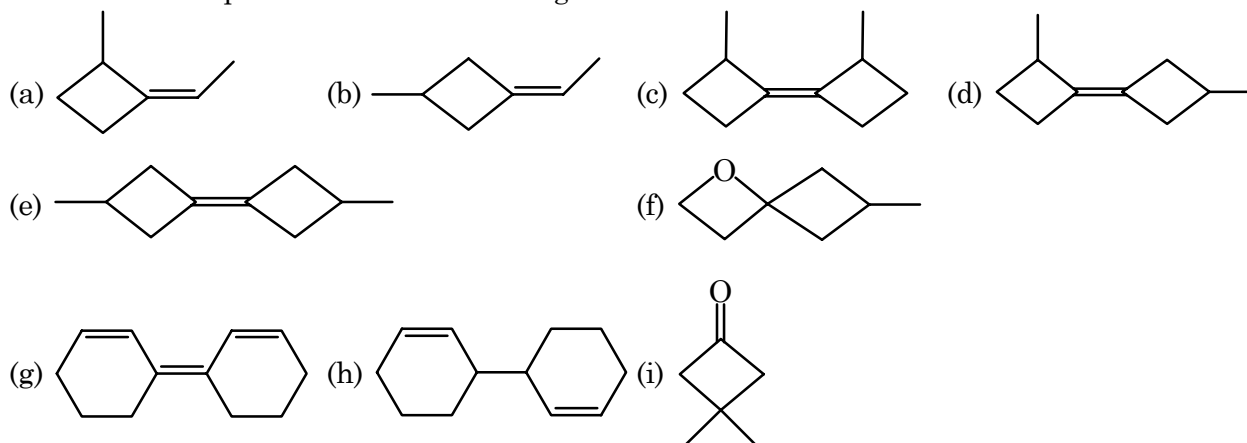
- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

- How many compounds are polar, planar and central atom is sp^3d hybridised
 (a) PCl_3F_2 (b) PF_3Cl_2 (c) XeF_2 (d) ClF_3
 (e) BrF_5 (f) I_3^- (g) SF_4 (h) S_3O_9
- One litre of a sample of hard water contain 2.22 mg $CaCl_2$ and 1.9 mg of $MgCl_2$. What is the total hardness in terms of ppm of $CaCO_3$. [Ca = 40 ; Mg = 24 , Cl = 35.5]
- How many of the following will show resonance



Space for Rough Work

4. Out of molecules CH_2Cl_2 , XeO_4 , BeCl_2 , XeO_2F_2 , BF_3 , NO_2^+ find total number of species which have similar hybridisation of the central atom as that of CO_2
5. Two bulbs A & B are emitting monochromatic light of wavelength such that A can just ionise H atom & B can just ionise He^+ ions. If the power of A & B are 30W & 40W respectively. The ratio of number of photons emitted per second by bulb A to bulb B is x : 1. The value of x is.
6. Number of compound which can show geometrical isomer



Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. The equation $x^3 - \frac{3}{4}x = -\frac{\sqrt{3}}{8}$ is satisfied by
(A) $x = \cos\left(\frac{5\pi}{18}\right)$ (B) $x = \cos\left(\frac{7\pi}{18}\right)$ (C) $x = \cos\left(\frac{23\pi}{18}\right)$ (D) $x = \cos\left(\frac{17\pi}{18}\right)$
2. The sum of the first three terms of an A.P. is 9 and the sum of their squares is 35. The sum of first n terms of the series can be
(A) $n(n+1)$ (B) n^2 (C) $n(4-n)$ (D) $n(6-n)$
-

Space for Rough Work

3. If the angles subtended by the sides of a triangle at orthocentre and incentre are equal, then the triangle is
- (A) Scalene
 - (B) Isosceles but not equilateral
 - (C) Equilateral
 - (D) Acute angled
4. Let $P(x) = ax^2 + bx + c$, $Q(x) = ax^2 + cx + b$ and $R(x) = ax^2 + bcx + b^3 + c^3 - 4abc$ where $a, b, c \in \mathbb{R}$ and $a \neq 0$. The equation $R(x) = 0$ will have non real roots if
- (A) $P(x) = 0$ has distinct real roots and $Q(x) = 0$ has non-real roots.
 - (B) $P(x) = 0$ has non-real roots and $Q(x) = 0$ has distinct real roots.
 - (C) Both $P(x) = 0$ and $Q(x) = 0$ have non-real roots.
 - (D) Both $P(x) = 0$ and $Q(x) = 0$ have distinct real roots.

Space for Rough Work

5. If real numbers x and y satisfy $(x + 5)^2 + (y - 12)^2 = (14)^2$, then the minimum value of $\sqrt{x^2 + y^2}$ is
- (A) rational (B) irrational
(C) integral (D) prime
6. Consider the parabola whose equation is $y = x^2 - 4x$ and the line $y = 2x - b$. Then which of the following is/are correct?
- (A) For $b = 9$ the line is a tangent to the parabola.
(B) For $b = 7$ the line cuts the parabola in A and B such that the $\angle AOB$ is a right angle when 'O' is the origin.
(C) For some $b \in \mathbb{R}$ the line cuts the parabola in C and D such that x-axis bisects the $\angle COD$.
(D) For $b > 9$ the line passes outside the parabola.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the line $x - y - 1 = 0$ intersect the parabola $y^2 = 8x$ at P & Q. The point of intersection of tangents at P & Q is (a, b) then $a + b$ is
2. Sum of the value(s) of λ such that the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$.
3. The maximum value of the function $f(x) = (7 \cos x + 24 \sin x)(7 \sin x - 24 \cos x)$, for every $x \in \mathbb{R}$.

Space for Rough Work

4. Consider a sequence $\{a_n\}$ with $a_1 = 2$ and $a_n = \frac{a_{n-1}^2}{a_{n-2}}$ for all $n \geq 3$, terms of the sequence being distinct. Given that a_2 and a_5 are positive integers and $a_5 \leq 162$ then sum of possible value(s) of a_5 is
5. The area of the triangular region in the first quadrant bounded on the left by the y-axis, bounded above by the line $7x + 4y = 168$ and bounded below by the line $5x + 3y = 121$, is
6. If lines $2x + 3y = 10$ and $2x - 3y = 10$ are tangents at the extremities of its same latus rectum to an ellipse whose centre is origin and axes are co-ordinate axes, then the length of the latus rectum is

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Number of solution(s) of the equation, $\tan 2x = \cot x$ in $0 \leq x \leq 2\pi$, is
2. Find the largest natural number 'a' for which the maximum value of $f(x) = a - 1 + 2x - x^2$ is smaller than the minimum value of $g(x) = x^2 - 2ax + 10 - 2a$.
3. If α, β are the roots of the quadratic equation $x^2 - \left(3 + 2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}}\right)x - 2\left(3^{\log_3 2} - 2^{\log_2 3}\right) = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2$ is equal to

Space for Rough Work

4. Let $A(2, -3)$ and $B(-2, 1)$ be vertices of a ΔABC . If the centroid of ΔABC moves on the line $2x + 3y = 1$, then the locus of the vertex C is $2x + 3y = \lambda$, then λ is
5. Maximum value of determinant $\begin{vmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{vmatrix}$ is
6. Let a_1, a_2, a_3, a_4 and b be real numbers such that $b + \sum_{K=1}^4 a_K = 8$ and $b^2 + \sum_{K=1}^4 a_K^2 = 16$, then the sum of possible positive integral value(s) of 'b' is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator